

Dataset Analysis for Decision-Making

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 sicor_municipal_co... .csv
16.7 MB

Analyze the provided dataset to identify its key characteristics for data-driven decision-making. Upon receiving a specific analytical question, address it strictly using the dataset. Your deliverables must include a clear, concise answer to the question and the Python code that extracts and processes the data used to derive your answer.

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To begin analyzing the dataset and addressing your request, I will first load the `sicor_municipal_contracts.csv` file and examine its structure, key characteristics, and contents. This initial exploration will help identify patterns, relationships, and potential insights that can support data-driven decision-making. Once the dataset's characteristics are understood, I will proceed to address any specific analytical question you provide.

Below is the Python code to load and explore the dataset:

Step 1: Load and Explore the Dataset

```
python
1 import pandas as pd
2
3 # Load the dataset
4 file_path = 'sicor_municipal_contracts.csv'
5 data = pd.read_csv(file_path)
6
7 # Display basic information about the dataset
8 print("Dataset Overview:")
9 print(data.info())
10
11 # Display the first few rows of the dataset
12 print("\nFirst 5 Rows of the Dataset:")
13 print(data.head())
14
15 # Summary statistics for numerical columns
16 print("\nSummary Statistics:")
17 print(data.describe())
18
19 # Check for missing values
20 print("\nMissing Values per Column:")
21 print(data.isnull().sum())
```

Explanation of the Code:

- Loading the Dataset :** The `pd.read_csv()` function reads the CSV file into a Pandas DataFrame.
- Basic Information :** The `data.info()` method provides an overview of the dataset, including column names, data types, and non-null counts.
- Preview Data :** The `data.head()` function displays the first five rows of the dataset, giving a glimpse of the data structure.
- Summary Statistics :** The `data.describe()` function computes descriptive statistics (e.g., mean, min, max) for numerical columns.
- Missing Values :** The `data.isnull().sum()` checks for missing values in each column, which is crucial for data cleaning.

Key Characteristics Identified:

After running the above code, I would analyze the output to identify:

- The number of rows and columns in the dataset.
- The data types of each column (e.g., numeric, categorical, datetime).
- Presence of missing or null values.



- Range and distribution of numerical variables.
- Unique values in categorical columns.

Next Steps:

Once you provide a specific analytical question, I will use the insights from this exploration to craft a targeted analysis. For example:

- If the question involves identifying trends, I will aggregate and visualize relevant data.
- If the question involves relationships between variables, I will compute correlations or group data accordingly.

Please provide your analytical question, and I will proceed with the detailed analysis and corresponding Python code.

Does the funding area impact the total contract value?



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To address the question of whether the funding area impacts the total contract value, we need to analyze the relationship between these two variables in the dataset. Here's how we can approach this:

Steps to Analyze the Relationship:

- 1. Identify Relevant Columns :**
 - The "funding area" column represents the categorical variable.
 - The "total contract value" column represents the numerical variable.
- 2. Group Data by Funding Area :**
 - Compute descriptive statistics (e.g., mean, median, sum) for the total contract value within each funding area.
- 3. Visualize the Relationship :**
 - Use a bar chart or box plot to visualize the distribution of total contract values across different funding areas.
- 4. Statistical Testing (Optional) :**
 - Perform an ANOVA test to determine if there are statistically significant differences in total contract values across funding areas.

Below is the Python code to perform this analysis:

Step 2: Analyze the Impact of Funding Area on Total Contract Value

```
python
1 import pandas as pd
2 import matplotlib.pyplot as plt
3 import seaborn as sns
4
5 # Load the dataset
6 file_path = 'sicor_municipal_contracts.csv'
7 data = pd.read_csv(file_path)
8
9 # Check column names to ensure correct references
10 print("Column Names:", data.columns)
11
12 # Group data by 'funding_area' and compute summary statistics for 'total contract value'
13 summary_stats = data.groupby('funding_area')['total_contract_value'].agg(['mean', 'median', 'sum'])
14 print("\nSummary Statistics by Funding Area:")
15 print(summary_stats)
16
17 # Visualize the mean total contract value by funding area
18 plt.figure(figsize=(10, 6))
19 sns.barplot(data=summary_stats, x='funding_area', y='mean', palette='viridis')
20 plt.title('Mean Total Contract Value by Funding Area')
21 plt.xlabel('Funding Area')
22 plt.ylabel('Mean Total Contract Value')
23 plt.xticks(rotation=45)
24 plt.tight_layout()
25 plt.show()
26
27 # Optional: Box plot to visualize the distribution of total contract values
28 plt.figure(figsize=(10, 6))
29 sns.boxplot(data=data, x='funding_area', y='total_contract_value', palette='coolwarm')
30 plt.title('Distribution of Total Contract Value by Funding Area')
31 plt.xlabel('Funding Area')
```

```
32 plt.ylabel('Total Contract Value')
33 plt.xticks(rotation=45)
34 plt.tight_layout()
35 plt.show()
```

Explanation of the Code:

- 1. Grouping and Aggregation :**
 - The `groupby()` function groups the data by the "funding area" column.
 - The `agg()` function computes the mean, median, sum, and count of the "total contract value" for each funding area.
- 2. Visualization :**
 - A bar plot shows the mean total contract value for each funding area, providing a clear comparison.
 - A box plot visualizes the distribution of total contract values within each funding area, highlighting variability.
- 3. Output :**
 - The summary statistics table provides numerical insights into the relationship.
 - The plots offer visual confirmation of any trends or differences.

Hypothetical Results and Interpretation:

Assuming the dataset contains meaningful data:

- If the mean total contract value varies significantly across funding areas, it suggests that the funding area does impact the total contract value.
- If the box plots show overlapping distributions with similar medians and interquartile ranges, it indicates little to no impact.

For example:

- If "Education" has a much higher mean total contract value compared to "Infrastructure," this would suggest that the funding area influences the contract value.
- Conversely, if all funding areas have similar means and distributions, the funding area may not be a significant factor.

Next Steps:

Run the code above to generate the results. If you provide the actual column names or additional details about the dataset, I can refine the analysis further.

Let me know if you'd like me to proceed with statistical testing (e.g., ANOVA) or additional analyses!

proceed



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To further analyze whether the funding area significantly impacts the total contract value, we can perform an **ANOVA (Analysis of Variance)** test. This statistical test determines if there are statistically significant differences in the means of total contract values across different funding areas.

Steps to Perform ANOVA:

- 1. Assumptions of ANOVA :**
 - Independence: Each observation should be independent.
 - Normality: The total contract values within each funding area should follow a normal distribution.
 - Homogeneity of Variance: The variance of total contract values should be similar across funding areas.
- 2. Prepare Data :**
 - Extract total contract values for each funding area into separate groups.
- 3. Perform ANOVA :**
 - Use Python's `scipy.stats.f_oneway()` function to compute the ANOVA test statistic and p-value.
- 4. Interpret Results :**
 - If the p-value is less than the significance level (e.g., 0.05), we reject the null hypothesis and conclude that the

- If the p-value is less than the significance level (e.g., 0.05), we reject the null hypothesis and conclude that the funding area has a significant impact on the total contract value.

Step 3: Perform ANOVA Test

Below is the Python code to conduct the ANOVA test:

```
python
1 import pandas as pd
2 from scipy.stats import f_oneway
3
4 # Load the dataset
5 file_path = 'sicor_municipal_contracts.csv'
6 data = pd.read_csv(file_path)
7
8 # Check column names to ensure correct references
9 print("Column Names:", data.columns)
10
11 # Drop rows with missing values in 'total_contract_value' or 'funding_area'
12 data_clean = data.dropna(subset=['total_contract_value', 'funding_area'])
13
14 # Extract total contract values for each funding area
15 funding_areas = data_clean['funding_area'].unique()
16 groups = [data_clean[data_clean['funding_area'] == area]['total_contract_value'] for area in funding_areas]
17
18 # Perform ANOVA test
19 f_stat, p_value = f_oneway(*groups)
20
21 # Print results
22 print("\nANOVA Test Results:")
23 print(f"F-statistic: {f_stat}")
24 print(f"P-value: {p_value}")
25
26 # Interpretation
27 if p_value < 0.05:
28     print("\nConclusion: The funding area has a statistically significant impact on the total contract value.")
29 else:
30     print("\nConclusion: There is no statistically significant impact of funding area on the total contract value.")
```

Explanation of the Code:

1. Data Cleaning :

- Rows with missing values in the "total_contract_value" or "funding_area" columns are dropped to ensure valid comparisons.

2. Grouping by Funding Area :

- The `data_clean.groupby('funding_area')` groups the data by funding area.
- A list comprehension extracts the total contract values for each funding area into separate groups.

3. ANOVA Test :

- The `f_oneway()` function computes the F-statistic and p-value for the groups.
- The null hypothesis assumes that all group means are equal.

4. Interpretation :

- A p-value < 0.05 indicates that at least one funding area has a significantly different mean total contract value compared to others.

Example Output:

Assuming the dataset contains meaningful data, the output might look like this:

```
1 Column Names: ['contract_id', 'funding_area', 'total_contract_value', ...]
2
3 ANOVA Test Results:
4 F-statistic: 12.45
5 P-value: 0.00012
6
7 Conclusion: The funding area has a statistically significant impact on the total contract value.
```

Visualizing Assumptions:

To validate the assumptions of ANOVA, we can create additional plots:



1. Normality Check :

```
python
1 import matplotlib.pyplot as plt
2 import seaborn as sns
3
4 # Plot histograms for each funding area
5 v for area in funding_areas:
6     sns.histplot(data_clean[data_clean['funding_area'] == area]['total_contract_value'], kde=True)
7 plt.title('Distribution of Total Contract Values by Funding Area')
8 plt.xlabel('Total Contract Value')
9 plt.ylabel('Frequency')
10 plt.legend()
11 plt.show()
```

2. Homogeneity of Variance :

```
python
1 import scipy.stats as stats
2
3 # Levene's test for homogeneity of variance
4 levene_stat, levene_p = stats.levene(*groups)
5 print("\nLevene's Test Results:")
6 print(f"Test Statistic: {levene_stat}")
7 print(f"P-value: {levene_p}")
8
9 v if levene_p < 0.05:
10     print("Warning: Variances are not homogeneous. Consider using a non-parametric test.")
11 v else:
12     print("Variances are homogeneous. ANOVA assumptions are satisfied.")
```

Next Steps:

1. Run the ANOVA test code to determine statistical significance.
2. Validate assumptions using the provided plots and tests.
3. If assumptions are violated, consider using a non-parametric alternative like the Kruskal-Wallis test.

Let me know if you'd like me to proceed with any additional analyses or visualizations!

